Data Preprocessing on the Mall Customer Dataset

Cleaning, Transformation, Dimensionality Reduction & Similarity Analysis

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1. Introduction & Objective

ABSTRACT

This report documents a complete data preprocessing pipeline applied to a raw mall / e-commerce customer dataset of **2,200 records across 25 attributes**. The raw data suffered from missing values, duplicate rows, inconsistent categorical labels, mixed date formats, and numerical outliers. A six-stage cleaning process reduced the data to **2,136 consistent records**, after which transformation, dimensionality-reduction, and similarity techniques prepared it for downstream analysis and modelling.

The objective of this assignment is to demonstrate a reproducible, end-to-end preprocessing workflow on a realistically "dirty" dataset. The work is organised into four phases: (i) data quality assessment, (ii) data cleaning, (iii) feature transformation and encoding, and (iv) dimensionality reduction and record-similarity analysis. Every step was implemented in Python using `pandas`, `numpy`, and `scikit-learn`, and is reproducible from the accompanying notebook.

1.1 Dataset Description

The dataset captures customer profile, transaction, and behavioural attributes. The 25 columns fall into four broad groups:

- **Identity:** `CustomerID`, `CustomerName`.
- **Demographic:** Age, Gender, City, Country.
- **Financial & transactional:** AnnualIncome_INR, IncomeCurrency, AvgBasketValue_INR, TotalPurchases, OnlinePurchases, StorePurchases, LoyaltyPoints.
- **Behavioural & categorical:** SpendingScore, MembershipTier, VisitFrequency, PreferredCategory, PaymentMethod, DeviceType, SatisfactionRating, CouponUsed, ChurnNextMonth, plus join / last-purchase dates and email provider.

2. Data Quality Assessment

Before cleaning, each column was profiled for data type, missing-value count and percentage, unique values, duplicates, invalid (negative) entries, and outliers via the inter-quartile-range (IQR) method. Missingness was widespread: 18 of 25 columns contained null values, several exceeding 10% of records. The most affected columns are summarised below.

Attribute	Missing	Missing %	Attribute	Missing	Missing %
Country	362	16.5%	DeviceType	254	11.5%
IncomeCurrency	318	14.5%	MembershipTier	243	11.0%
CouponUsed	265	12.0%	Gender	230	10.5%
VisitFrequency	259	11.8%	EmailProvider	228	10.4%
PaymentMethod	220	10.0%	AnnualIncome_INR	141	6.4%
Age	104	4.7%	LastPurchaseDate	76	3.5%

The assessment also revealed **64 fully duplicated records** and pervasive inconsistency in categorical fields — e.g. Gender as M / F / male / Female; currency as Rs / INR / ₹; and multiple spellings of the same city. Date columns mixed formats (`24-Feb-2024`, `14/09/2024`, `2022-05-30`). The full profiling table is exported as `MallCustomer_DataQuality_Report.csv`.

3. Data Cleaning

Cleaning was performed in six sequential stages. After all stages the dataset contained **2,136 records × 25 columns** with no missing values in the imputed fields.

3.1 Missing-Value Imputation

Numerical columns were imputed with the **median** (robust to skew and outliers), and categorical columns with the **mode** (most frequent category), preserving each attribute's central tendency without distorting its distribution.

3.2 Duplicate Removal

Exact duplicate rows were dropped with `drop_duplicates()`, removing **64 records** (2,200 → 2,136). Duplicates inflate frequency counts and bias any trained model, so their removal is a prerequisite for reliable analysis.

3.3 Category Standardization

Whitespace was stripped from all categorical fields, then value-mapping dictionaries consolidated variant labels onto a canonical form. Representative mappings:

Attribute	Variant values (before)	Canonical (after)
Gender	`M`, `F`	Male, Female
IncomeCurrency	`Rs`, `Inr`, `₹`	INR
MembershipTier	`Silvr`	Silver
PreferredCategory	`Electronic`, `Eletronics`, `Groceries`	Electronics, Grocery
City	`Bangalore`, `Hyd`, `Chenai`, `Calcutta`	Bengaluru, Hyderabad, Chennai, Kolkata
PaymentMethod	`Card`, `Upi`	Credit Card, UPI
CouponUsed	`1/Y`, `0/N`	Yes, No

3.4 Data-Type Conversion

The JoinDate and LastPurchaseDate fields were parsed to datetime with `pd.to_datetime(errors='coerce', dayfirst=True)` so dates become sortable and support duration / recency features.

3.5 Noise Handling

A second whitespace-stripping pass over all text columns removed residual leading/trailing spaces (e.g. " gmail.com ", " bengaluru ") introduced during data entry.

3.6 Outlier Treatment (IQR Capping)

Outliers in nine numerical columns were treated by **IQR capping** (winsorization): values below `Q1 - 1.5·IQR` or above `Q3 + 1.5·IQR` were clipped to the respective bound. Capping was preferred over deletion so no records are lost while extremes are contained.

Column	Outliers capped	Column	Outliers capped
StorePurchases	142	AnnualIncome_INR	107
OnlinePurchases	140	Age	67
AvgBasketValue_INR	139	SatisfactionRating_1_5	28
TotalPurchases	109	LoyaltyPoints	17
SpendingScore_1_100	0	—	—

4. Data Transformation & Encoding

To make attributes comparable in scale and usable by distance-based and linear models, five transformation techniques were applied, each to a representative column:

Technique	Applied to	Purpose
Min–Max Normalization	AnnualIncome_INR	Rescales values to the [0, 1] range.
Standardization (Z-score)	AvgBasketValue_INR	Centres to mean 0, unit variance.
Log Transformation `log1p`	LoyaltyPoints	Compresses right-skew; safe at zero.
Label Encoding	CouponUsed	Maps binary category to 0 / 1.
One-Hot Encoding	MembershipTier	Expands nominal tiers to indicator columns.

Normalization and standardization prevent high-magnitude features (such as income) from dominating distance and gradient computations. The log transform reduces skew in the loyalty-points distribution, while label and one-hot encoding convert categorical fields into numeric form for machine-learning algorithms.

5. Dimensionality Reduction

5.1 Feature Selection

Non-predictive identifier and date columns (`CustomerID`, `CustomerName`, `JoinDate`, `LastPurchaseDate`) were removed as they carry no generalisable signal. The transformed frame then had **33 columns**, of which **22 were numerical** features fed into PCA.

5.2 Principal Component Analysis (PCA)

The 22 numerical features were standardized and reduced with PCA configured to retain **95% of variance**. PCA compressed the space to **16 principal components**, together explaining **95.07%** of total variance — an eight-feature reduction with negligible information loss. Leading components and their explained-variance ratios:

PC1	PC2	PC3	PC4	PC5	PC6	…	PC16	Cumulative
10.1%	9.2%	9.1%	7.8%	5.8%	5.4%	…	3.9%	**95.07%**

The relatively even spread of variance across components indicates the customer attributes are weakly correlated, so no single component dominates — a realistic property for behavioural data.

6. Record-Similarity Analysis

To illustrate distance and similarity measures, the first 20 customer records were standardized and compared using two metrics:

- **Euclidean distance** — straight-line distance in feature space; smaller means more alike.
- **Cosine similarity** — angle between feature vectors, in [−1, 1]; higher means more alike.

After excluding self-similarity, the most similar pair among the sample was **Record 7 and Record 12**, with a cosine similarity of **0.8531**. Such similarity structure is the basis of customer segmentation, recommendation, and nearest-neighbour approaches.

7. Conclusion

A raw, inconsistent dataset of 2,200 records was transformed into a clean, analysis-ready table of 2,136 records through systematic missing-value imputation, duplicate removal, categorical standardization, type conversion, noise handling, and outlier capping. Feature transformation and encoding placed all attributes on comparable numeric scales, PCA reduced 22 numerical features to 16 components while retaining 95% of variance, and a similarity analysis demonstrated how cleaned, standardized data supports downstream tasks. The pipeline is fully reproducible from the accompanying notebook, and the cleaned dataset is delivered as mall_customer_cleaned.csv.

Key Outcomes

Metric	Before	After
Records	2,200	2,136
Duplicate rows	64	0
Columns with missing values	18	0 (imputed)
Numerical features for modelling	22	16 (PCA, 95%)